Analysis: In this scenario, the interviewer is evaluating multiple skills:

- The ability to articulate the benefits of component-based design (e.g., modularity, reusability, scalability, separation of concerns, and encapsulation).
- Skill in decomposing a UI into logical, manageable parts (e.g., splitting BlogPostApp into FavoriteArticle, Like, and Comment sections).
- Understanding of stateful vs. stateless components and knowing when to use each.
- Familiarity with best practices for organizing components and feature-based folder structures.
- Practical knowledge of implementing and composing components, managing props and state, and leveraging React Hooks effectively.
- Awareness of trade-offs in design decisions (e.g., monolithic vs. modular components, and when class components may still be needed).
- The ability to communicate and justify design choices under time constraints.

Jane's Approach: She realized that a naive solution—a single, monolithic component handling all functionality—might work for small projects but leads to poor testability, maintenance issues, and limited reusability. To embrace React's philosophy, she suggested breaking the UI into smaller, reusable components. This modular design enhances flexibility and scalability, as shown in figure 4.2.

Figure 4.2 Principles of component-based architecture: modularity, reusability, and scalability